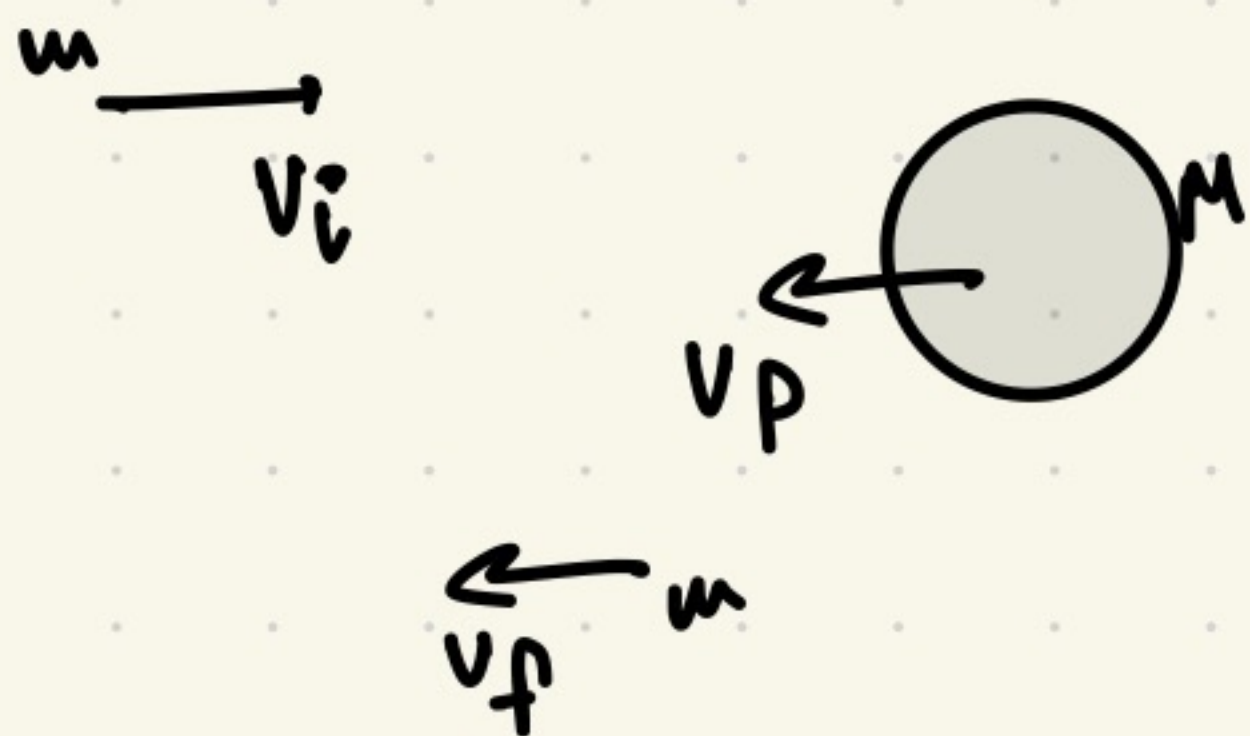


①



Assuming elastic collision

$$e = \frac{v_p + v_i}{v_f - v_p} = 1 \quad \left(\text{as } M \gg m, \text{ final velocity of the planet remains unchanged} \right)$$

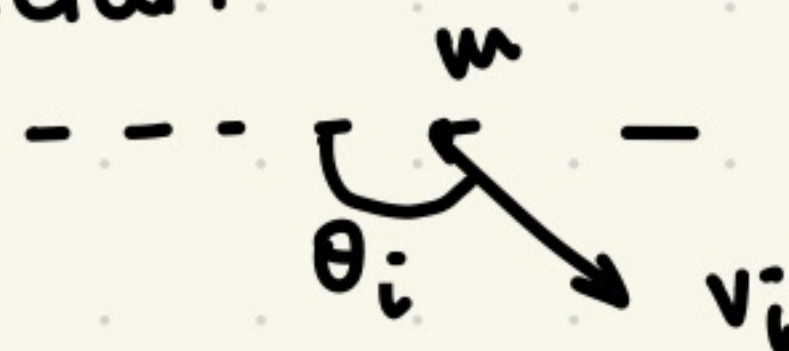
$$v_p + v_i = v_f - v_p$$

$$|v_f| = |v_i| + 2|v_p|$$

$$\bar{v}_p = 2\bar{v}_p - v_i$$

②

θ is the angle between the velocity of spacecraft and the planet's velocity.



$$|\bar{v}_i| = |\bar{v}_i - \bar{v}_p|$$

$$|v_i|^2 = |v_i|^2 + |v_p|^2 - 2|v_i||v_p|\cos\theta_i$$

$$|v_f|^2 = |v_f|^2 + |v_p|^2 - 2|v_f||v_p|\cos\theta_f$$

$$|v_i| = |v_f|$$

$$|v_i|^2 - 2|v_i||v_p|\cos\theta_i = |v_f|^2 - 2|v_f||v_p|\cos\theta_f$$

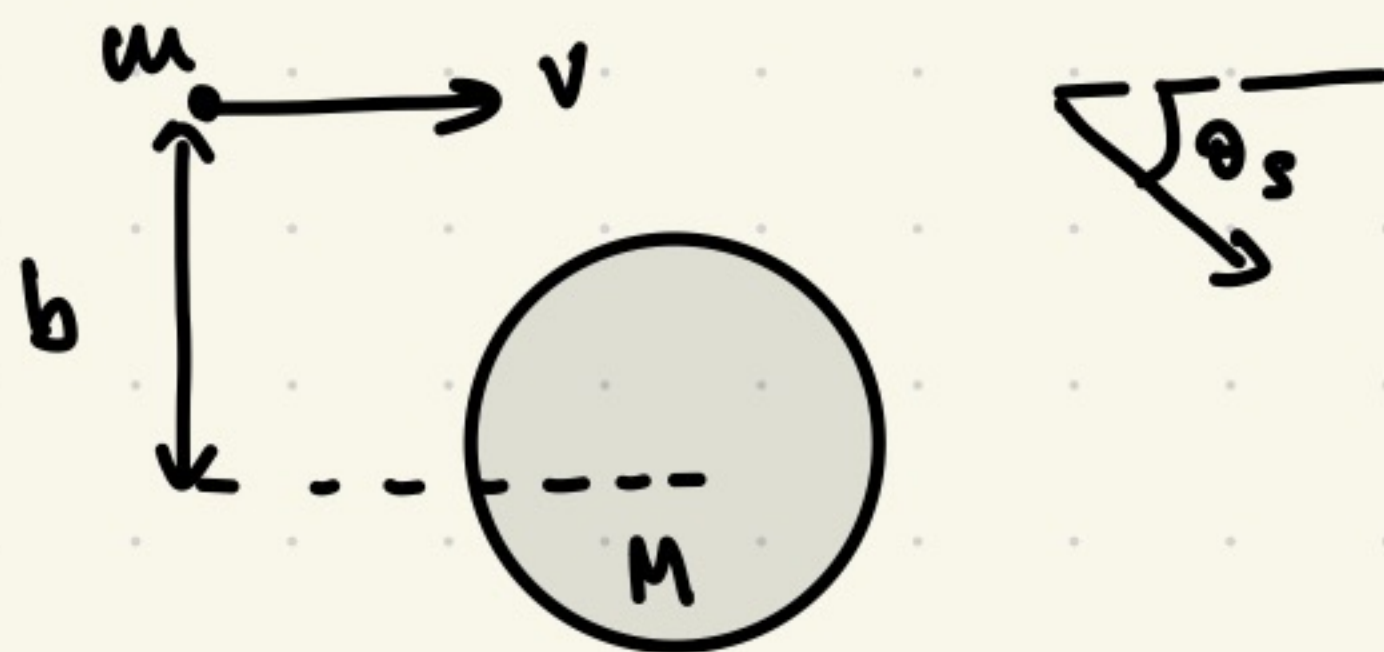
$$|v_f| = v$$

$$v^2 - 2v|v_p|\cos\theta_f + 2|v_i||v_p|\cos\theta_i - |v_i|^2 = 0$$

$$v = \frac{2|v_p|\cos\theta_f + \sqrt{4|v_p|^2\cos^2\theta_f + 4|v_i|^2 - 8|v_i||v_p|\cos\theta_i}}{2}$$

$$|v_f| = |v_p|\cos\theta_f + \sqrt{|v_p|^2\cos^2\theta_f + |v_i|^2 - 2|v_i||v_p|\cos\theta_i}$$

③



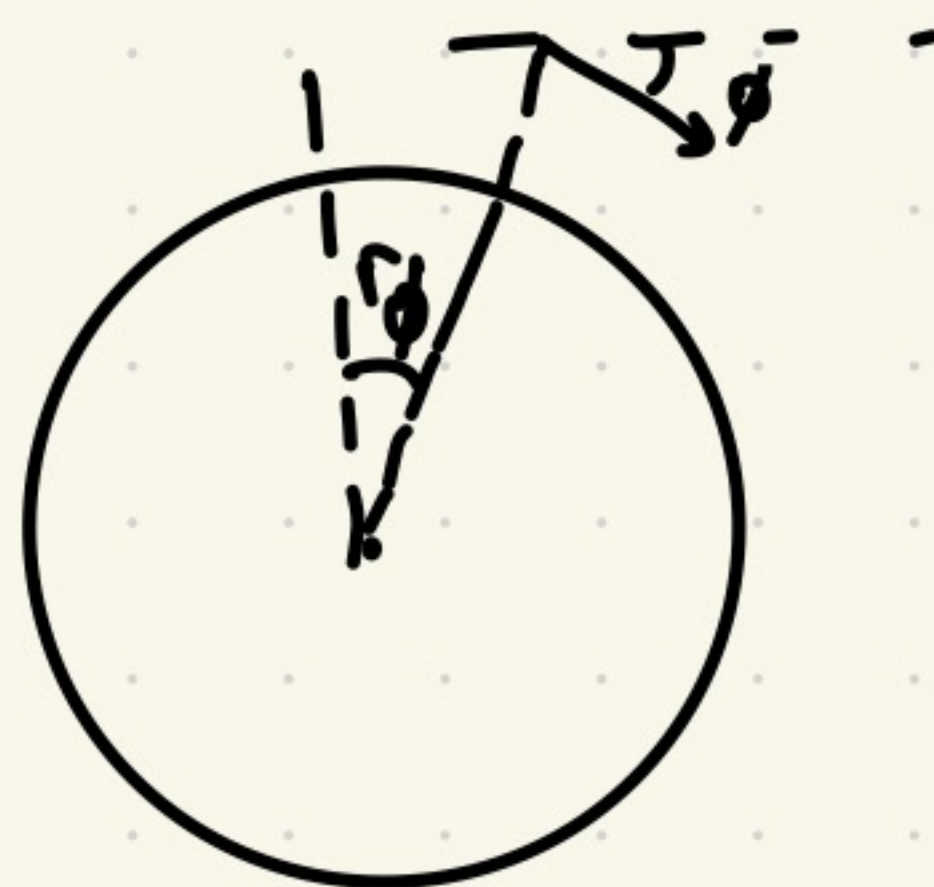
final speed = initial speed (cons. of Energy)

Approach-1 : I will first solve using mechanics, then by hyperbola properties

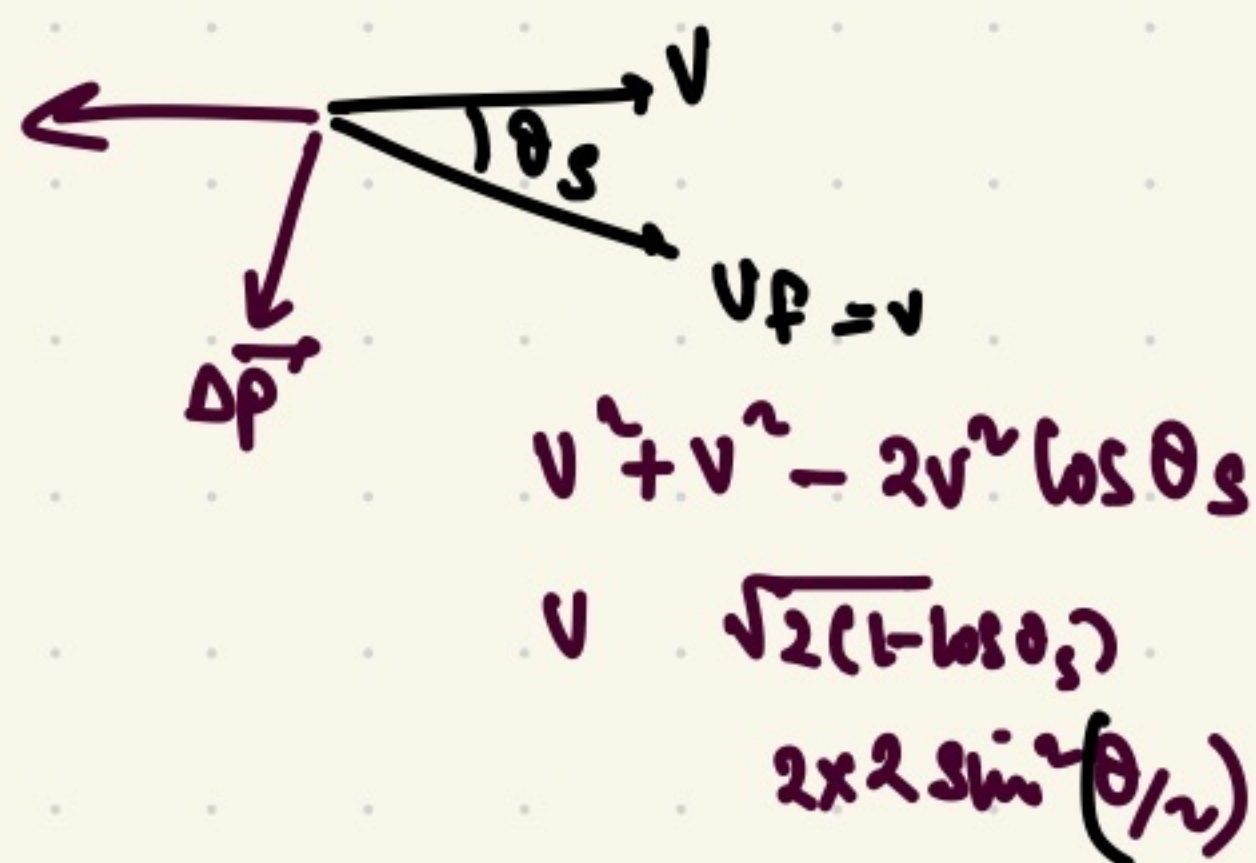
- Angular momentum is conserved.

$$mv_b = m \frac{d\phi}{dt} r^2$$

$$dt = \frac{r^2}{v_b} d\phi$$



- Change in momentum



$$\Delta \vec{p} = 2v \sin(\theta/2) m$$

$$2mv \sin \frac{\theta}{2} = \frac{GMm}{v_b} \times 2 \cos(\frac{\theta}{2})$$

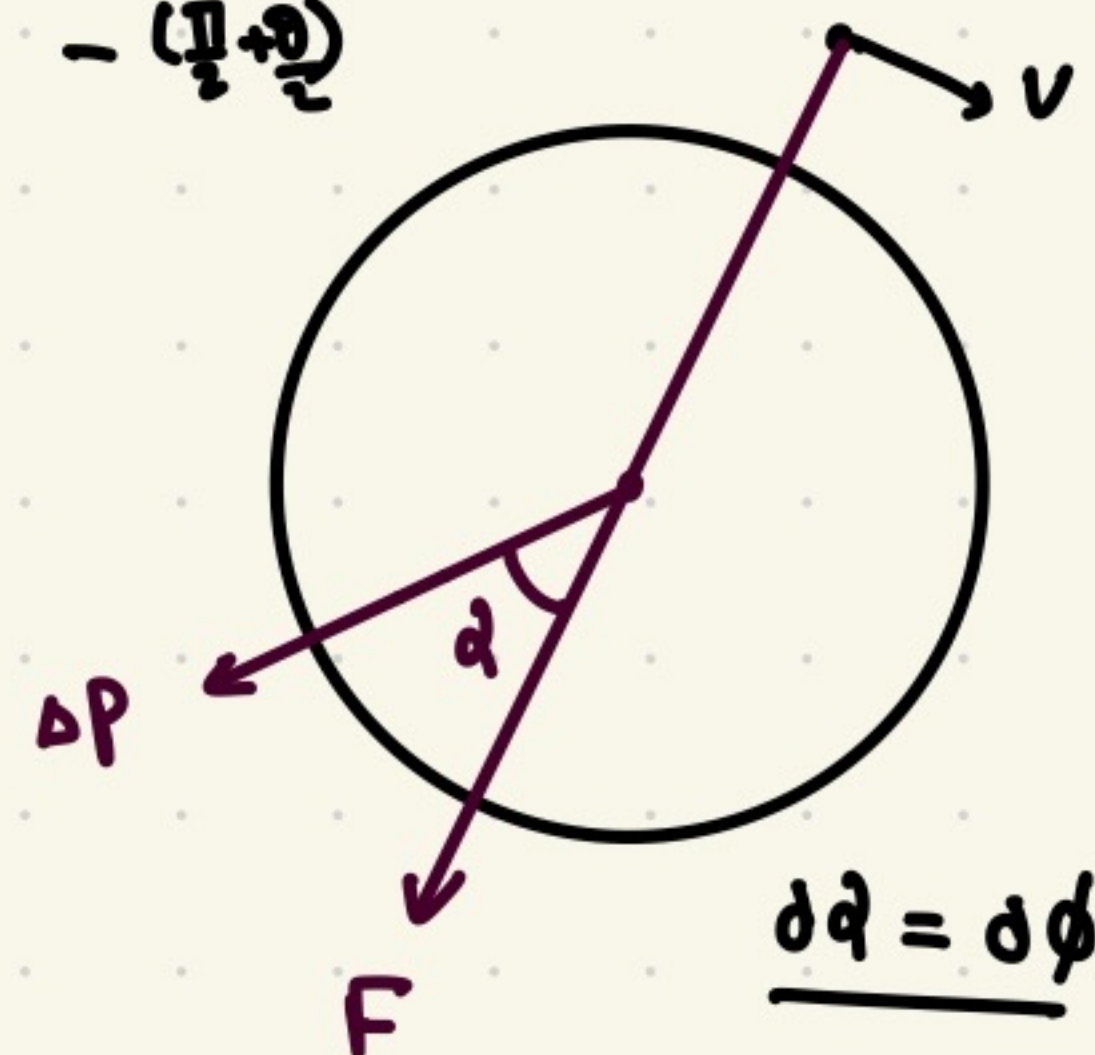
$$b = \frac{GM}{v^2_b} \cot(\theta/2)$$

$$\theta = 2 \tan^{-1} \left(\frac{GM}{v^2_b b} \right)$$

- Impulse due to gravitational force.

Since we know only the Impulse in the dirn of $\Delta \vec{p}$ will survive

$$I = \int_{-\frac{\theta}{2}}^{+\frac{\theta}{2}} F \cos \alpha d\alpha$$



$$I = \int \frac{GMm}{r^2} \times \cos \alpha \frac{r^2}{v_b} d\phi$$

$$= \frac{GMm}{v_b} [\sin \alpha]_{-\frac{\theta}{2}}^{+\frac{\theta}{2}}$$

$$I = \frac{GMm}{v_b} \times 2 \cos(\frac{\theta}{2})$$

Closest Approach.

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_i^2 - \frac{GMm}{r_0}$$

$$mvb = mv_i r_0$$

$$v_i = \frac{vb}{r_0}$$

$$r_0^2 + \frac{2bM}{v_0^2} r_0 - b^2 = 0 \Rightarrow$$

$$r_0 = \frac{bM}{v^2} \left(\left(1 + \left(\frac{bv^2}{GM} \right)^2 \right)^{1/2} - 1 \right)$$

Approach-2 Hyperbola Approach.

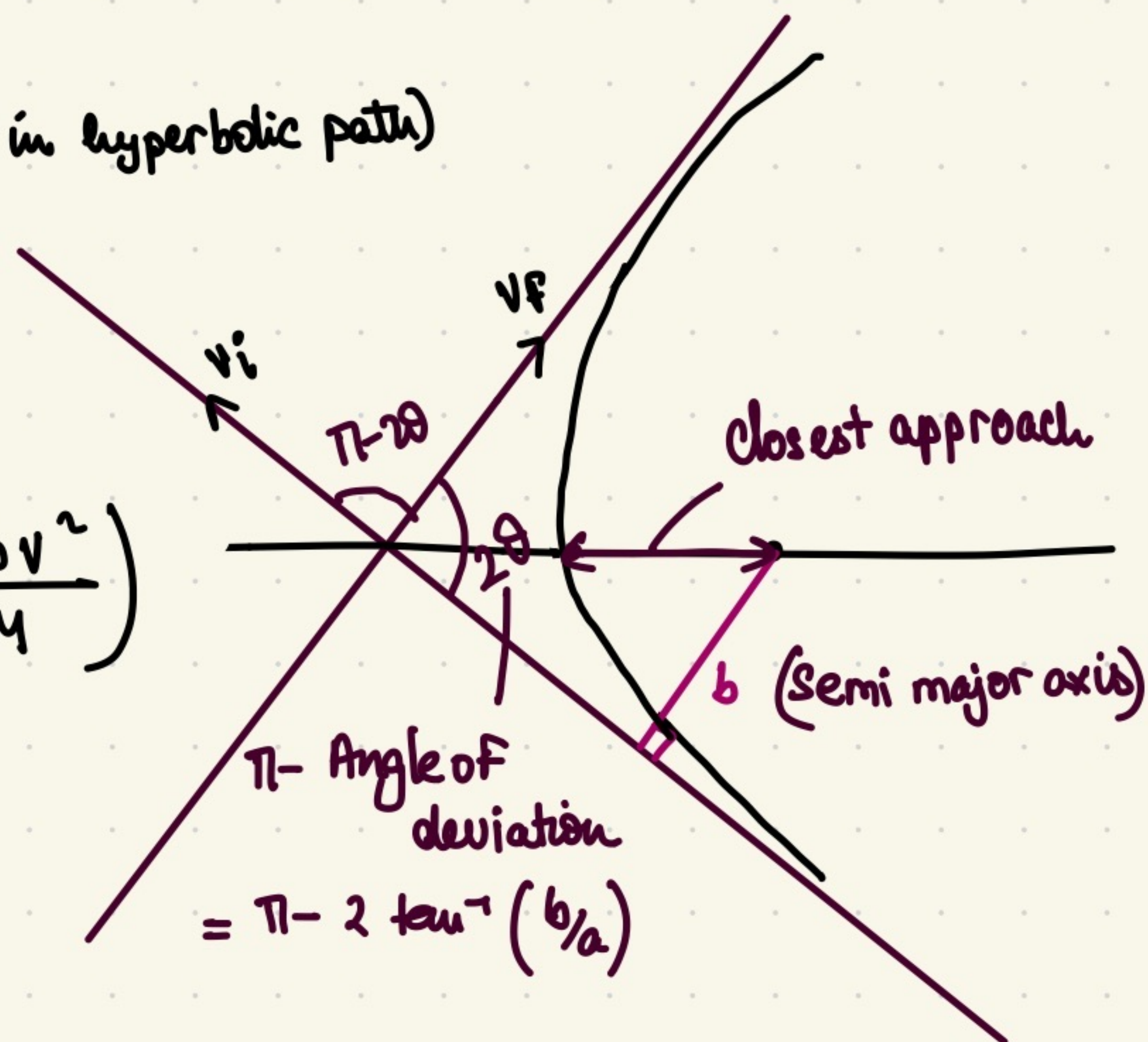
Energy Conservation :-

$$\frac{1}{2}mv^2 = \frac{GMm}{a} \quad (\text{Total Energy in hyperbolic path})$$

$$a = \frac{GM}{v^2}$$

$$\text{Angle of deviation} = \pi - 2 \tan^{-1} \left(\frac{bv^2}{GM} \right)$$

$$\theta_s = 2 \tan^{-1} \left(\frac{GM}{bv^2} \right)$$



$$\begin{aligned} \text{Closest approach} &= a(e-1) \\ &= a \left(\sqrt{1 + \frac{b^2}{a^2}} - 1 \right) \end{aligned}$$

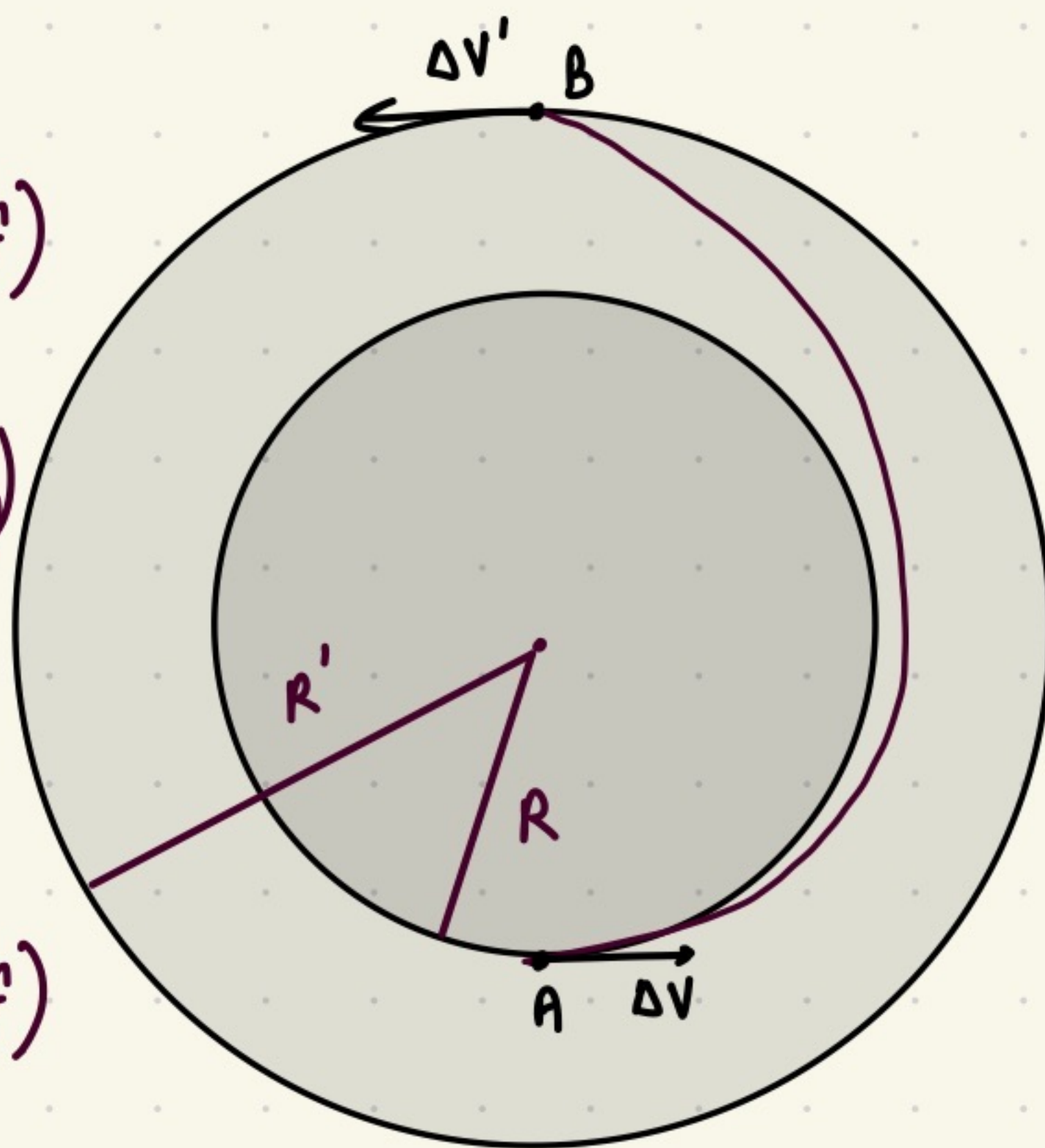
$$= \frac{GM}{v^2} \left(\left(1 + \left(\frac{bv^2}{GM} \right)^2 \right)^{1/2} - 1 \right)$$

④ (a) $(v_i)_{\text{orbit}_1} = \sqrt{\frac{GM}{R}} \quad \left(\frac{mv^2}{r} = \frac{GMm}{r^2} \right)$

$(v_f)_{\text{orbit}_1} = \sqrt{\frac{2GM}{(R+R')} \times \frac{R}{R}} \quad \left(\text{L and E conservation} \right)$

$(v_i)_{\text{orbit}_2} = \sqrt{\frac{2GM}{(R+R')} \times \frac{R}{R'}} \quad (1)$

$(v_f)_{\text{orbit}_2} = \sqrt{\frac{GM}{R'}} \quad \left(\frac{mv^2}{r} = \frac{GMm}{r^2} \right)$



$$\Delta V = \sqrt{\frac{GM}{R}} \left(\sqrt{\frac{2R'}{R+R'}} - 1 \right)$$

$$\Delta V' = \sqrt{\frac{GM}{R'}} \left(1 - \sqrt{\frac{2R}{R+R'}} \right)$$

(b) Time in ellipse, $T^2 = \frac{4\pi^2 a^3}{GM}$

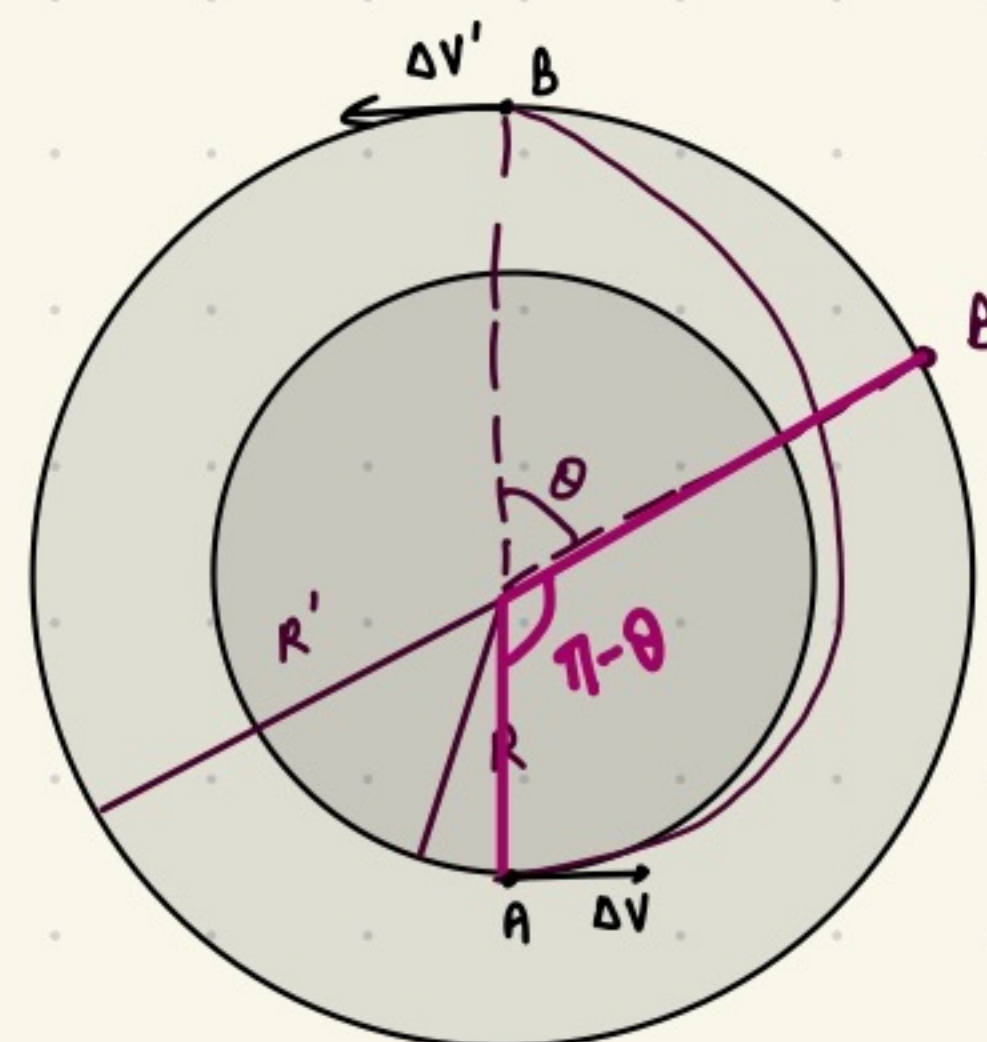
$a = \frac{R+R'}{2} \Rightarrow T = \frac{2\pi \left(\frac{R+R'}{2} \right)^{3/2}}{(GM)^{1/2}}$

Time taken from pt A to B = $\frac{T}{2} = \frac{\pi \left(\frac{R+R'}{2} \right)^{3/2}}{(GM)^{1/2}}$

(c) Jupiter covers θ in Time the satellite approaches B from A.

$$\frac{\theta(R)^{3/2}}{\sqrt{GM}} = \frac{\pi \left(\frac{R+R'}{2} \right)^{3/2}}{\sqrt{GM}}$$

$$\pi - \theta = \pi \left(1 - \left(\frac{1 + \frac{R'}{R}}{2} \right)^{3/2} \right)$$



5

$$v_p = \frac{2\pi R}{T} = 12.917 \text{ km/s}$$

$$\omega_p = 0.523 \text{ rad/yr}$$

$$v_s = 36000 \text{ km/h} = 10 \text{ km/s}$$

v_p



$$b = 6.5 \times 10^8 \text{ m}$$

v_{rel} to planet

$$v_{rel} = \left(|v_p|^2 + |v_s|^2 - 2|v_p||v_s|\cos 60^\circ \right)^{1/2}$$

$$= 11.73 \text{ km/s} = 11730 \text{ m/s}$$

(a)

$$\text{distance of closest approach} = \frac{GM}{v^2} \left(\left(1 + \left(\frac{v^2 b}{GM} \right)^2 \right)^{1/2} - 1 \right)$$

$$= 9.21 \times 10^8 \left(2.24 \times 10^{-1} \right)$$

$$= 2.062 \times 10^8 \text{ m}$$

(b)

$$\text{Angle of deviation} = 2 \tan^{-1} \left(\frac{GM}{bv^2} \right)$$

$$= 2 \tan^{-1} (1.417)$$

$$= 109.57^\circ$$

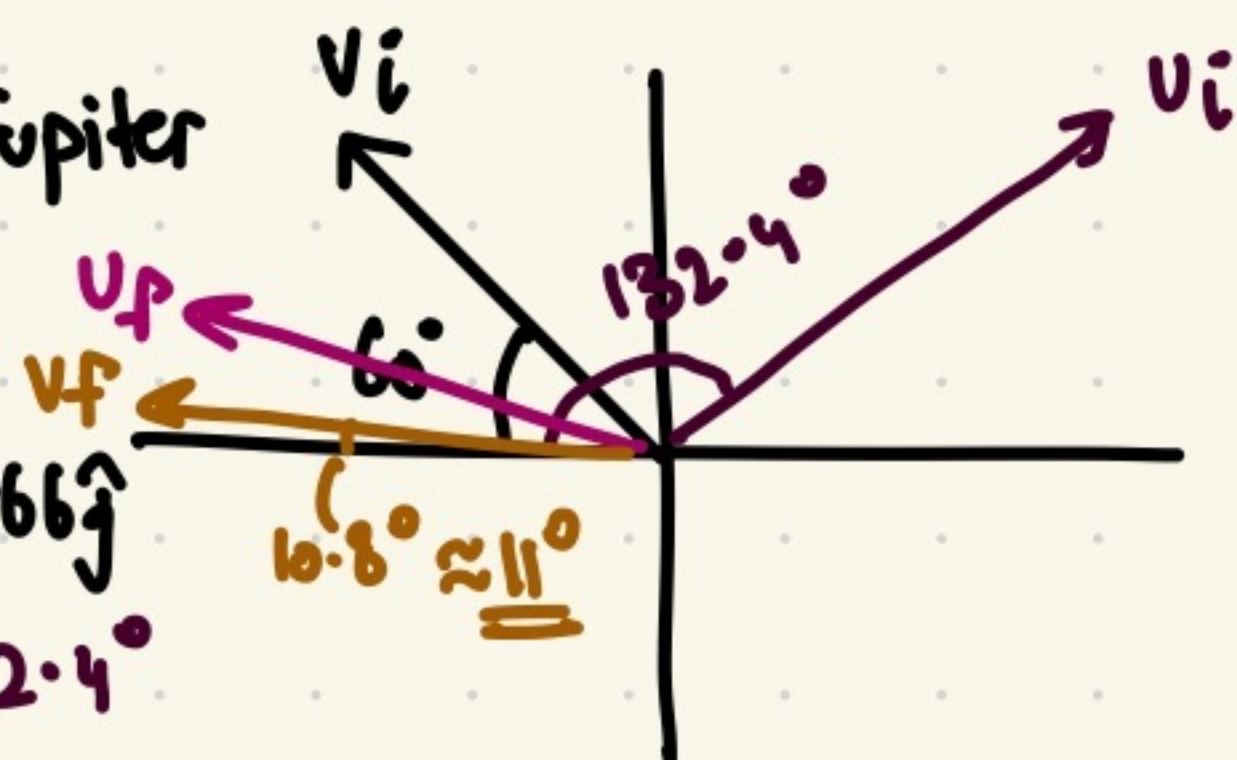
Since, the satellite goes to neptune, I will assume gravitational assist.

• Angle in frame of jupiter

$$\vec{v}_i = \vec{v}_i - \vec{v}_p$$

$$= (5 + 12.917)\hat{i} + 8.66\hat{j}$$

$$\tan^{-1} \left(\frac{8.66}{7.917} \right) = 132.4^\circ$$

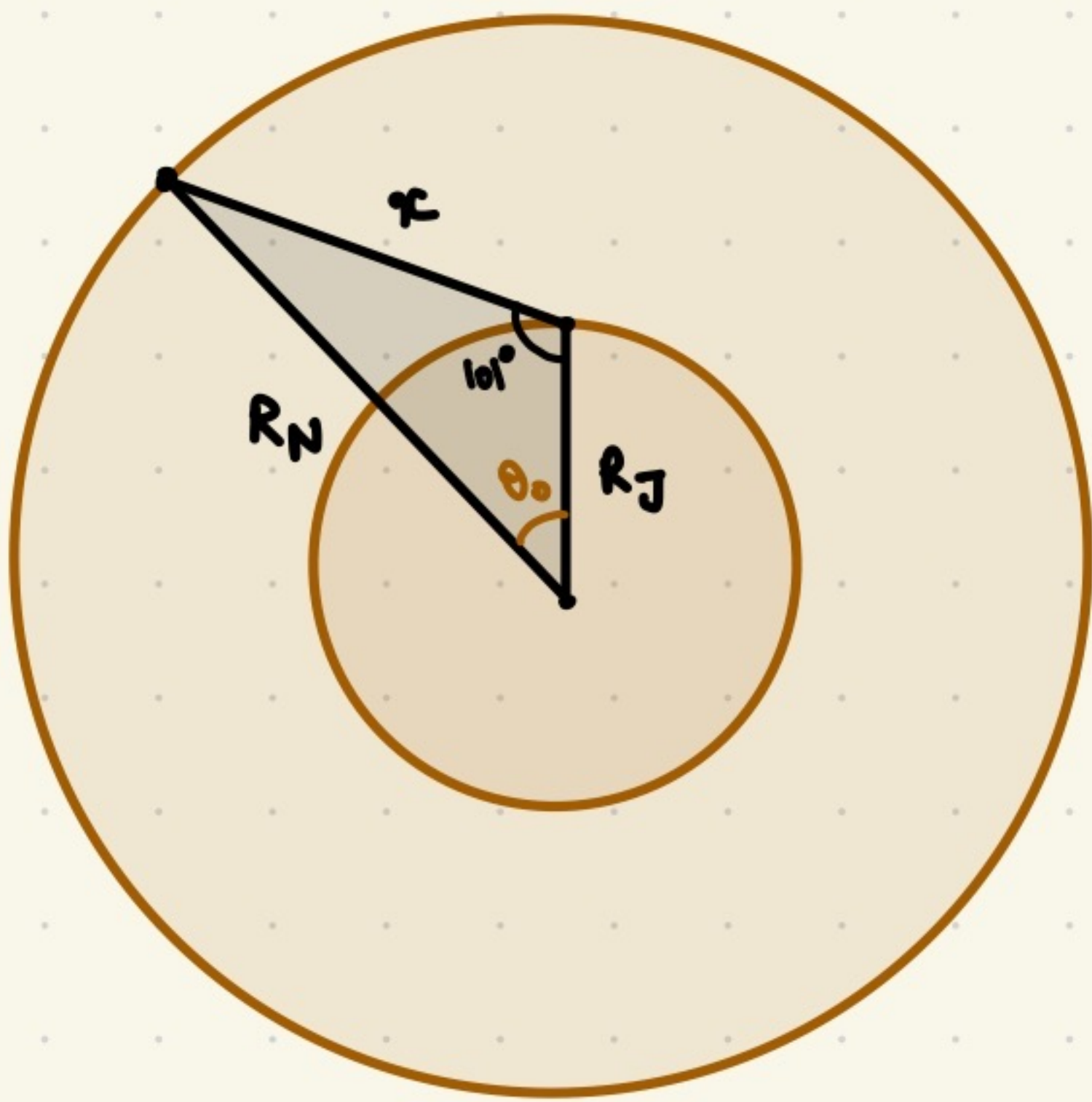


$$\text{final Angle } v_f = 132.4^\circ - 109.57^\circ = 22.8^\circ$$

$$\vec{v}_f = \vec{v}_i + \vec{v}_p$$

$$(-10.81 - 12.917)\hat{i} + 4.54\hat{j}$$

$$\theta = 10.8$$



$$|\vec{v}_f| = 24.1 \text{ km/s}$$

$$\cos 101^\circ = \frac{x^2 + R_J^2 - R_N^2}{2 x R_J}$$

$$= x^2 - 2 \cos 101^\circ R_J x + R_J^2 - R_N^2 = 0$$

$$x = \frac{2 \cos 101^\circ R_J \pm \sqrt{(2 \cos 101^\circ R_J)^2 + R_J^2 - R_N^2}}{2}$$

$$x \approx -0.19 R_J + \sqrt{R_N^2 - R_J^2}$$

$$x \approx 4.32 \times 10^{12} \text{ m}$$

$$\text{Time taken to cover} = \frac{4.32 \times 10^9 \text{ km}}{24.1} \text{ sec}$$

$$= 5.68 \text{ yrs.}$$

$$\omega_J = 0.523 \text{ rad/yr}$$

$$\omega_N = 3.81 \times 10^{-2} \text{ rad/yr}$$

Kepler's law

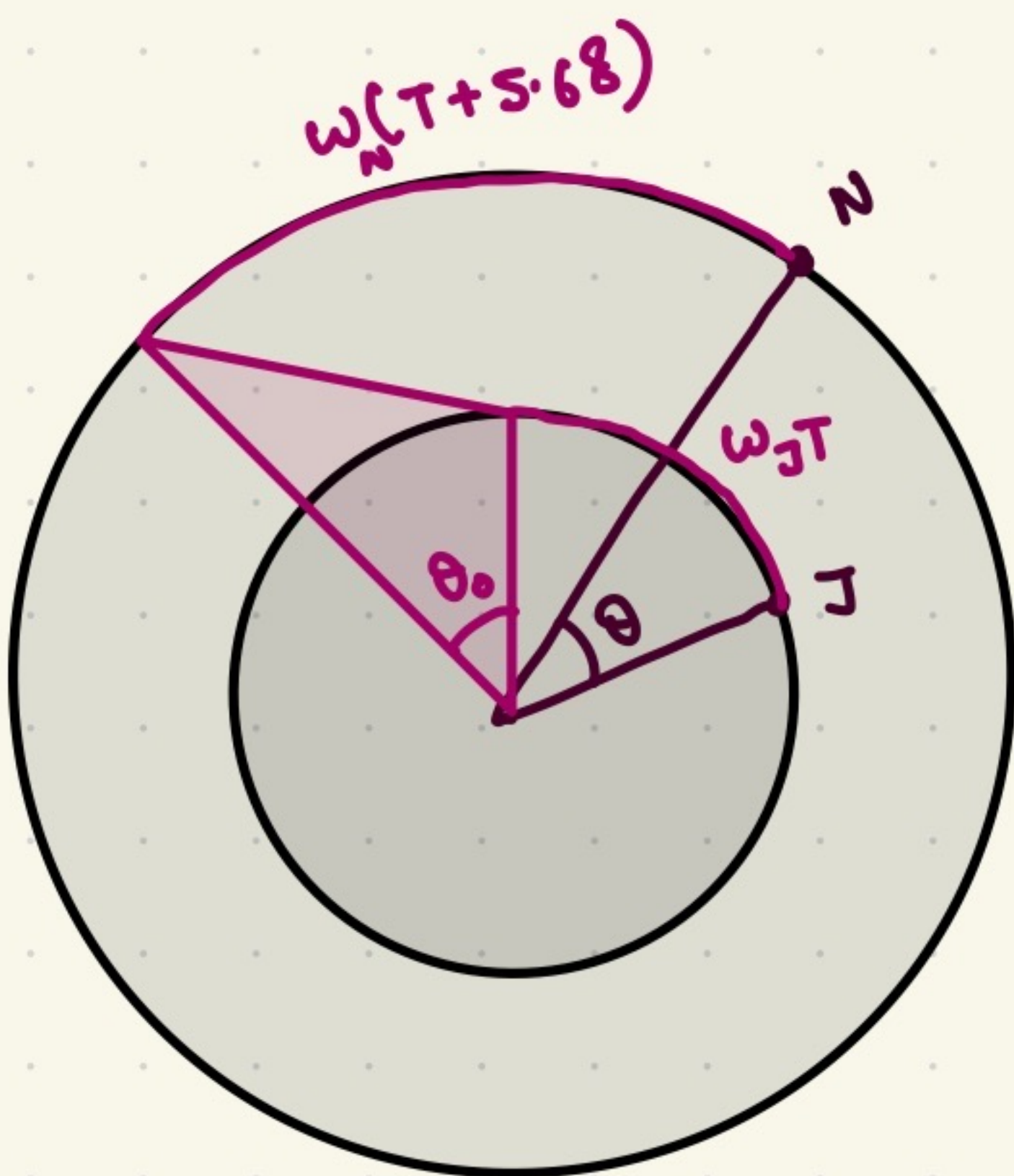
$$T^2 \propto R^3$$

$$\frac{T_J^2}{T_E^2} = \frac{R_J^3}{R_E^3}$$

$$R_E = \frac{R_J}{(12)^{2/3}} = 1.48 \times 10^{11} \text{ m}$$

$$T = \frac{\pi \left(\frac{R_E + R_J}{2} \right)^{3/2}}{\sqrt{GM_S}} = 2.72 \text{ yrs}$$

$$(M_S = 2 \times 10^{30}) \quad (T - \text{Time from earth to jupiter})$$



$$\theta + \omega_N(T + 5.68) - \omega_J(T) = \theta_0$$

$$\theta = \theta_0 + (\omega_J - \omega_N)T - \omega_N(5.68)$$

$$= 1.21 + 0.484 T - 0.216$$

$$\theta = 1.42 + 0.484 T \quad \text{Radians}$$

$$= 1.42 + 0.484 \times 2.72$$

$$= 2.736 = \underline{\underline{156.78^\circ}}$$